Pavanbabu Arjunamahanthi

Gainesville, FL, USA

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Education

Doctor of Philosophy in Electrical and Communications Engineering **Jan 2025 – May 2026**
University of Florida Gainesville, FL
CGPA: 3.62/4

Master of Science in Materials Engineering **Aug 2023 – May 2025**
University of Florida Gainesville, FL
CGPA: 3.51/4

Bachelor of Technology in Materials Engineering **Oct 2019 – Jun 2023**
National Institute of Technology, Srinagar Srinagar, India
CGPA: 7.98/10

Technical Skills

Programming: Python, MATLAB
Software: Thermo-Calc, Origin, CrystalMaker, Python, Figma, KLayout,
Characterization: Atomic Force Microscopy (AFM), X-Ray Diffraction, X-ray Microscopy, Scanning Acoustic Microscopy, Scanning Electron Microscopy, Focused Ion Beam, EDS, Dilatometry, XPS, XRF, Raman Spectroscopy, FTIR, TGA, DSC, Ellipsometry, UV–Vis, Profilometry, IV/CV Measurement, 4-point probe Measurement, Vibrating Sample Magnetometry
Fabrication: Photolithography, Deep Reactive Ion Etching, Evaporation, Plasma Enhanced Chemical Vapor Deposition, Doping, Diffusion, ALD

Research & Work Experience

Graduate Research Assistant **January 2025 - December 2025**
Advisor: Navid Asadi Zanjani, SCAN LAB, Department of ECE, **University of Florida** Gainesville, FL

- **Focus:** Physical Inspection of Electronics, Glass Interposers, Photonic Integrated Circuits, Hybrid Bonding, Advanced Packaging, MEMS, Semiconductor Fabrication

Research Intern **Dec 2022 – Feb 2023**
Advisor: Nand Kishore Prasad, Department of Metallurgical Engineering, **IIT-BHU** Varanasi, India
Project: *Synthesis of rare-earth-free nanocomposites for magnetic and alternative energy applications*

- Achieved **95% yield** synthesizing and characterizing high entropy alloy nanoparticles via high-energy (cryo) ball milling.
- Engineered nanocomposites for solid-state batteries using sol–gel and co-precipitation methods with heat treatments; increased material purity and structural homogeneity by **30%**.
- Published: “Magnetic properties of hard-soft MnBi/FeNi@C exchange coupled nanostructures via cryo-milling.”

Research Projects

1. PSM: Ushering PCB Security Through System-level MEMS Integration **Jan 2025 – Dec 2025**
Advisor: Navid Asadi Zanjani

- Co-developed MEPS tool, a MEMS-based circuit obfuscation framework introducing reconfigurable routing within PCBs to prevent reverse engineering and Counterfeiting attacks.
- Demonstrated resistance against imaging and probing attacks, including X-ray tomography, SAM, and short test probing.
- Achieved 100% non-destructive inspection goals with X-ray, SAM, and Optical Microscopy of fabricated MEMS Devices by proving resolution limitations towards reverse engineering, contributing to secure and reliable Defense-grade electronics.

2. Semiconductor Device Fabrication Project **Jan 2025 – May 2025**

- Fabricated a PN junction Diode and a Bi-Morph MEMS Actuator by literature-backed parameter optimizing on a Wafer by Photolithography, HF Wet and Dry Etching, Diffusion, Doping, Evaporation, and Plasma Cleaning, saving 20% cost and resources.
- Demonstrated hands-on parallel working proficiency in microfabrication tools, PECVD, DRIE, ALD, and metrology tools such as Ellipsometry, Profilometry, and SEM Analysis, with experience in Wafer handling, cutting iterative processes by 30%.
- Conducted device characterization with IV/CV, 4-Probe Measurement, Static and Dynamic Resonance of MEMS, Semiconductor Parameter Analyzer, and ensured 70% working devices by careful fabrication.

2. Hybrid Bonding Interface and Reliability Characterization **Feb 2025 – Present**
Advisor: Navid Asadi Zanjani

- Developed a reproducible metallographic workflow for sub-10 μm hybrid bonding, improving SEM cross-sectional fidelity for reliability-critical failure analysis.

- Validated on Cu–Cu and SiO_2–SiO_2 interfaces, enabling precise detection of voids, delamination, and misalignment.
- Engineered sub-5 micrometer hybrid bonding cross-sectional imaging by using targeted ultrasonication and HF Etching, without FIB delayering, improving SEM Quality, reducing charging and contamination artifacts by 99% and preparation cost reduction by over 50%.
- Developed a new sub-surface interfacial taxonomy with 20+ hybrid bonding manufacturing defects for AI-driven metrology and correlated subsurface defects to warpage behavior, leading to process development.

3. Scanning Acoustic Microscopy (SAM) Imaging and Microbump Fingerprinting **May 2025 – Present**
Advisor: Navid Asadi Zanjani

- Collaborated and developed the **first SAM-based fingerprinting method** for microbumps in packages by analyzing 450+ images across 8 samples under varied thermal/imaging conditions, contributing towards future secure electronic packages.
- Identified voiding and delamination in advanced packages to train **Physics-informed Neural Networks** for defect detection and imaging optimization.
- Accomplished a SAM-based microbump fingerprinting security approach by automating the testing process, segmenting, and statistically analyzing over 20000 microbumps across 450 SAM images taken in 8 Advanced packages, saving 40% project time.
- Collaborated with **SkyWater Technology** for SAM-based post-bonding testing of hybrid and fusion-bonded wafers.

4. Virtualized Training in Advanced Packaging **Aug 2024 – Aug 2025**
Advisors: Navid Asadi Zanjani, Hyo J. Kang

- Co-designed ChipXR, a **GPT-powered XR lab** for Advanced Packaging education, enabling immersive, cost-effective training.
- Validated models for microbumps, TSVs, and die geometries for 100% technical accuracy; deployed in classroom settings; contributed to **Meta Quest Store** release and design recommendations.

5. Multi-Scale Probing for Advanced Packaging Systems Vulnerability **Jan 2024 – Aug 2024**
Advisor: Victoria Miller

- • Led material characterization work on 2.5D/3D advanced packaging interconnects, identified 10+ material-related failures and probing regions in 4+ package components and 10+ package interfaces, and developed strong probing countermeasures.
- Performed materials characterization for root-cause analysis at TSV/interposer/microbump levels (e.g., Kirkendall voids, bridging, intermetallic growth).

Publications

Journal Articles

- Anuraag, N. S., **A. Pavan Babu**, and N. K. Prasad. **“Magnetic properties of hard-soft MnBi/FeNi@ C exchange coupled nanostructures via cryo-milling.”** Physica B: Condensed Matter 693 (2024): 416411.

Conferences

- **Arjunamahanthi, Pavanbabu**, et al. **“Revitalizing Advanced Packaging in America.”** 2025 Pan Pacific Strategic Electronics Symposium (Pan Pacific). IEEE, 2025.
- Kottur, H. R., M. S. M. Khan, **Pavanbabu Arjunamahanthi**, L. K. Biswas, N. Varshney, L. S. Hayes, B. Liu, M. Delany, J. Hihath, N. AsadiZanjani. **“MEMS-Enabled Secure PCB Supply Chain.”** In 2025 IEEE WMED, pp. 1–4. IEEE, 2025.
- **Arjunamahanthi, Pavanbabu**, et al. **“Interactive Learning in Microelectronics Education: Comparing PC and Mixed Reality Approaches for Student Engagement and Visual-Spatial Memory.”** CHI EA 2025.
- Biswas, L. K., H. R. Kottur, **Pavanbabu Arjunamahanthi**, H. Wang, H. Dalir, N. AsadiZanjani. **“Hardware Fingerprinting for SiPh Assurance by Metasurface-based Density-Controlled Patterns.”** IEEE RAPID 2025.
- **Arjunamahanthi, Pavanbabu**, et al. **”A Global View Versus a US Focus on Outsourced Assembly and Test (OSAT) Facilities to Support Wafer Level Packaging.**” Wafer-Level Packaging Symposium. Vol. 21. No. 1. Surface Mount Technology Association, 2025.
- H. R. Kottur, **Pavanbabu Arjunamahanthi**, M. S. M. Khan, L. K. Biswas, N. Varshney, N. AsadiZanjani. **“NEMS for Hardware Security in Advanced Packaging.”** iMAPS 2025.
- **Arjunamahanthi, Pavanbabu**, Himanandhan Reddy Kottur, Shajib Ghosh, Patrick Craig, M. Shafkat M. Khan, Liton Kumar Biswas, Istiaq Firoz Shiam, Navid Asadizanjani, Robert Patti, and Charles Woychik. **“The Future of Electronics Packaging Is Chiplet Architecture.”** In SMTA International, vol. 27, no. 1, pp. 82-92. Surface Mount Technology

Association, 2025.

- **Pavanbabu Arjunamahanthi**, H. R. Kottur, S. Ghosh, P. Craig, M. S. M. Khan, L. K. Biswas, S. I. Firoz, N. AsadiZanjani, R. Patti, C. Woychik. **“The Future of Electronics Packaging Is Chiplet Architecture”** Poster presented at iMAPS ThermoCon-Semiconductor Thermal Management Conference 2025, San Diego, USA.
- H. R. Kottur, **Pavanbabu Arjunamahanthi**, L. K. Biswas, S. I. Firoz, N. AsadiZanjani **“Inspection and Metrology Challenges in Hybrid Bonding Manufacturing,”** Published in ASM-EDFA Magazine for May 2, 2026 Issue.
- Kottur, Himanandhan Reddy, **Pavanbabu Arjunamahanthi**, Nusra Akte Takia, Istiaq Firoz Shiam, Navid Asadizanjani, Victor Vilar, Robert Patti, and Charles Woychik. **”An Update on the Global View versus a US Focus of Outsourced Assembly and Test (OSAT) Facilities to Support Wafer Levl Packaging.”** In Wafer-Level Packaging Symposium, vol. 22, no. 1, pp. 10-19. Surface Mount Technology Association, 2026.

Accepted Conferences for Paper Submission

- L. K. Biswas, H. R. Kottur, S. I. Firoz, **Pavanbabu Arjunamahanthi**, N.Hubble, P.Handler, V.Vilar, D.Setiadi, C.G.Woychik N. AsadiZanjani. **“Predictive Reliability Modeling of Hybrid Bonding through Warpage and Interfacial Defect Correlation”** Accepted for IEEE ECTC 2025, USA.
- H. R. Kottur, **Pavanbabu Arjunamahanthi**, M. S. M. Khan, P.A.Titty, L.S.Hayes, L. K. Biswas, K.Yahyaei, B.Liu, S. I. Firoz, M.Delany, J.Hihahth, N. Asadizanjani. **“SUBSTRATE-CLOAK: A MEMS-Enabled Reconfigurable Obfuscation Framework for Secure Substrate Interconnects”** Accepted for IEEE ECTC 2025, USA.

Scholarly Contributions

Technical Contributor — Introduction to the Microelectronics Physical Assurance Book (Springer)

- Contributed technical diagrams, editing, and content review for the book “Introduction to Microelectronics Advanced Packaging Assurance,” with special mention received in the foreword for technical accuracy and formatting.

Reviewer Activity

- Reviewed two manuscripts for **Springer Nature** and **IEEE TCAD**.

Video Contributor – Advanced Packaging Educational Series on YouTube

- Co-created and presented technical video content on Advanced Packaging modules, targeting industry and academic audiences.
- Collaborated with Dr. Nourouddin Sharifi’s team at **Tarleton State University** and **ficonTEC**. Published on Navid Asadi Y.T
- Topics Published on YouTube: Advanced Packaging Thermal Management Series **(17 Videos)**, Optical Transceivers in Silicon Photonics (**1 Video**)

Leadership

Physics Mentor — STEP-UP 2024 **June 2024 – August 2024**

- Mentored 47 undergrads, improved physics learning performance by 37% by redesigning classroom teaching with real-life problems, and by implementing the flipped classroom model. .

Awards and Recognition

- University of Florida **Academic Achievement Award** **Aug 2023 – Dec 2024**
- Featured my research paper **”The Future of Electronics packaging is Chiplet Architecture”** on prominent semiconductor news sites such as Semiconductor Digest, Semi Vision, Chiplet Marketplace, and 3D InCites.
- Featured my Co-authored paper **”Nanoelectromechanical Systems (NEMS) for Hardware Security in Advanced Packaging”** on semiengineering website, including archival in its repository.
- Received the **“Best Practical and Applications-Based Knowledge”** category award at SMTA International 2025 for my first-authored paper **”The Future of Electronic Packaging is Chiplet Architecture”**.